



Computer Networks CL3001

LAB – 06(A) Telnet, SSH

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
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Computer Networks Lab 06

Course: Computer Networks (CL3001)

Semester: Spring 2025

Instructor: Muhammad Nadeem Ghouri

Note:

- Maintain discipline during the lab.
 - Listen and follow the instructions as they are given.
 - Just raise hand if you have any problem.
 - Completing all tasks of each lab is compulsory.
 - Get your lab checked at the end of the session.
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Lab Objective

- Introduction to Telnet & configuration of Telnet in Cisco Packet Tracer.
- Introduction to SSH & configuration of SSH in Cisco Packet Tracer.
- Introduction to ACL & configuration of ACL in Cisco Packet Tracer.

SSH & Telnet

1. Introduction to Telnet

A terminal emulation program for TCP/IP networks such as the Internet. The Telnet program runs on your computer and connects your PC to a server on the network. You can then enter commands through the Telnet program and they will be executed as if you were entering them directly on the server console. This enables you to control the server and communicate with other servers on the network. To start a Telnet session, you must log in to a server by entering a valid username and password. Telnet is a common way to remotely control Web servers.

To telnet means to establish a connection with the Telnet protocol, either with command line client or with a programmatic interface.

2. Configuration of Telnet

Below are the steps for Telnet Protocol. Follow the procedure for the configuration of Telnet Protocol.

Step 1: Build the following topology.

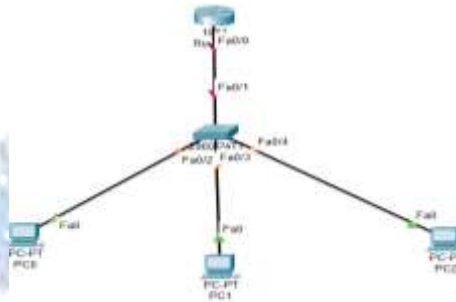


Figure 1

Step 2: Set IPs on the PCs. As, by default, all PCs are in vlan. We will create a virtual interface on switch with vlan 1 as follows:

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

Figure 2

Assigning IP to PCs:

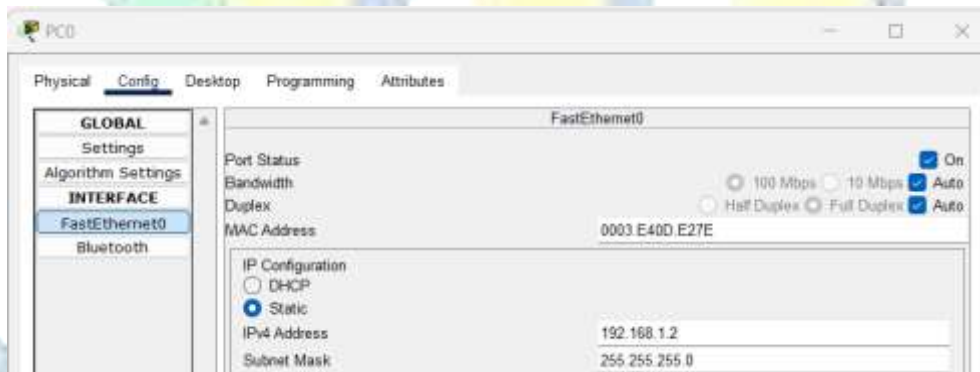


Figure 3



Figure 4



Figure 5

Step 3: Now, try to telnet the switch from our PC, it refuses because we have not applied authentication on the switch yet.

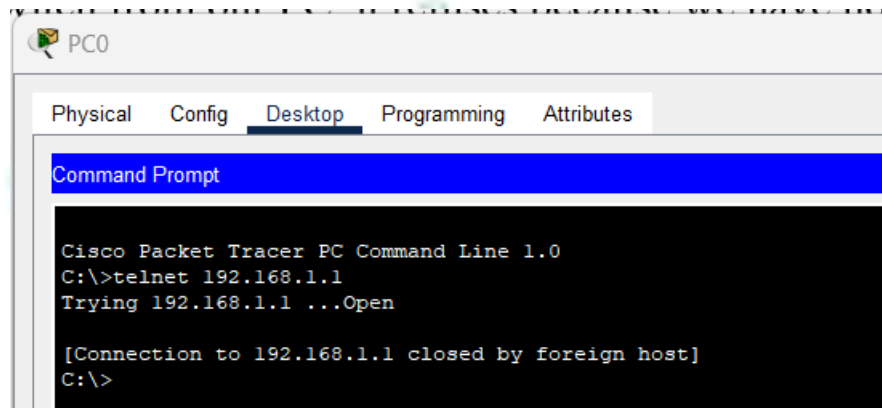


Figure 6

Secure Shell Server (SSH) and FTP services. Each Telnet, SSH, or FTP session requires one vty-line. You can add security to your system by configuring the software to validate login requests:

```
Switch(config-if)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
```

Figure 7

Now, we can easily telnet. But it does not let us go in the switch enabled mode because we have not set the password on the switch yet.

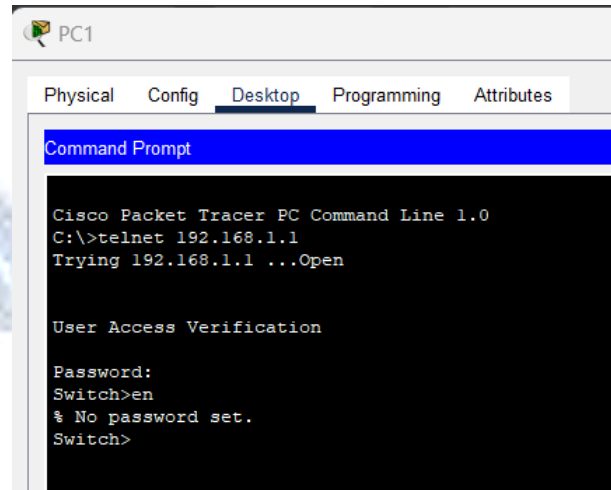


Figure 8

Let's apply password on the switch enabled mode.

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable password cs
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 9

Now, we can go inside Switch configuration mode from our pc.

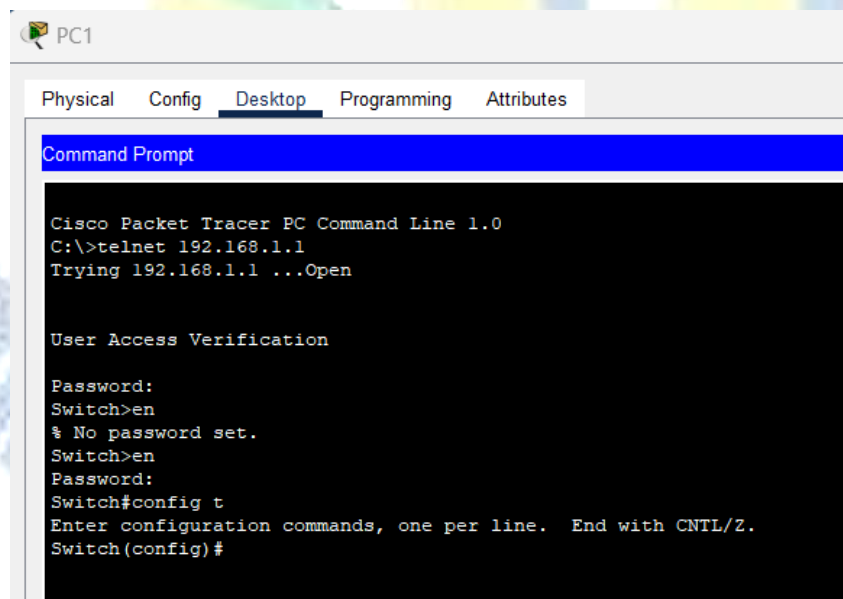


Figure 10



3. Introduction to SSH

Secure Shell or Secure Socket Shell is a network protocol. It is an application layer protocol that is in the 7th layer of the Open Systems Interconnection (OSI) network model. It also refers to the suite of utilities that implements the SSH protocol.

Secure Shell also supports both password and key-based authentication. Password-based authentication let users provide username and password to authenticate to the remote server.

A key-based authentication allows users to authenticate through a key-pair. The key pairs are two cryptographically secure keys for authenticating a client to a Secure Shell server.

Furthermore, the Secure Shell protocol also encrypts data communication between two computers. It is extensively used to communicate with a remote computer over the Internet.

4. Configuration of SSH

Taking the same topology as mentioned in telnet.

Below are the steps for SSH Protocol. Follow the following procedure for the configuration of SSH Protocol.

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname sameer
sameer(config)#ip domain name ai-study
sameer(config)#crypto key generate rsa
The name for the keys will be: sameer.ai-study
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
sameer(config)#
```

Figure 11

```
sameer(config)#ip s
*Mar 1 0:29:49.71: %SSH-5-ENABLED: SSH 1.99 has been enabled
% Ambiguous command: "ip s"
sameer(config)#ip ssh version 2
sameer(config)#line vty 0 15
sameer(config-line)#transport input ssh
sameer(config-line)#
```

Figure 12

Protocol working on it. By default, username is admin. We can apply any sort of configuration on our switch from our PC.



```
C:\>ssh -l admin 192.168.1.1
Password:
Password:

sameer>en
Password:
sameer#config t
Enter configuration commands, one per line. End with CNTL/Z.
sameer(config)#interface fa0/2
sameer(config-if)#no shutdown
sameer(config-if)#exit
sameer(config)#exit
sameer#
```

Figure 13

Now, if we want to change the username from admin to something else, we will do it as follows:

```
sameer(config)#username ds-study secret abc
sameer(config)#line vty 0 15
sameer(config-line)#login local
sameer(config-line)#
```

Figure 14

Now we will do the following on our PCs:

```
C:\>ssh -l ds-study 192.168.1.1
Password:
% Login invalid

Password:
% Login invalid

Password:

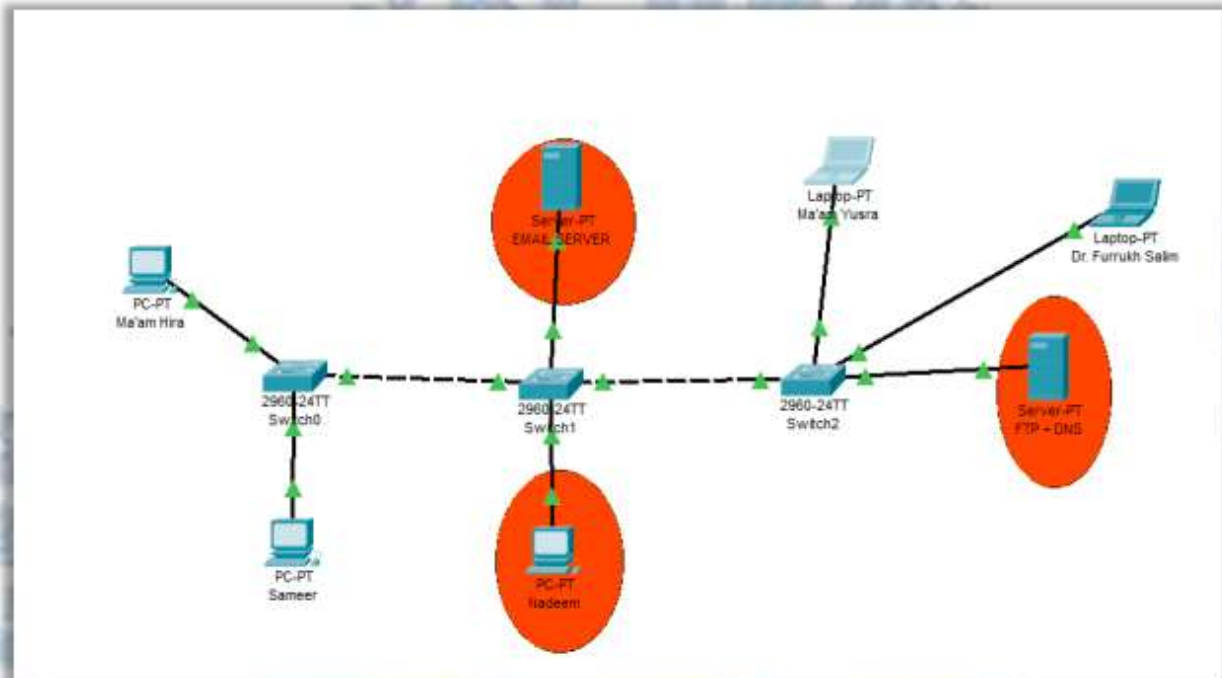
sameer>
```

Figure 15



Lab 06-A Tasks Report Submission

Task-01 Implement the topology given below on cisco packet tracer:

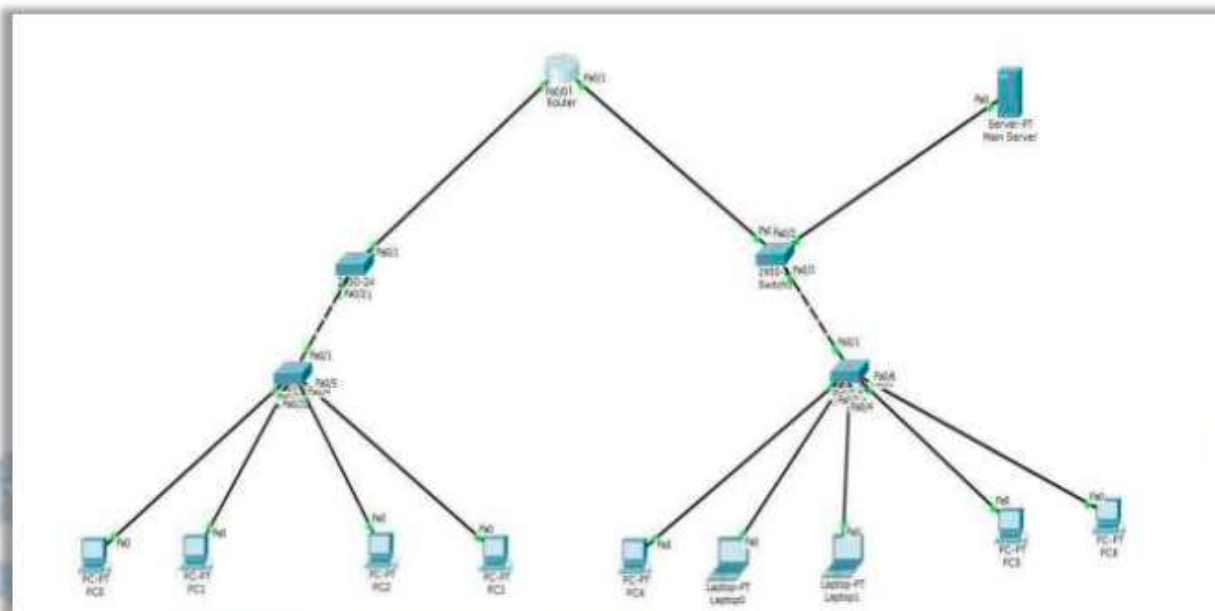


Do the following:

- Assign IP to the computers/Laptops. The Network should like this XX.XX.YY.0. i.e. your roll number like 3879(38.79.1.0) and for all other networks Y should be replaced by 2, 3 and so on.
- Ping the server from any computer.
- Verify the telnet connection from all switches nearest to the computer.
- Do change the IP of Switch2 from PC user Nadeem using its command prompt.



Task-02 Implement the topology given below on cisco packet tracer:



Do the following:

- The IPs should be assigned to the computer using static method and to the router using CLI. The Network on one side of FastEthernet should like this XX.XX.YY.0 i.e. your roll number like 3879(38.79.1.0) and on another side it should be 3880(38.80.2.0).
- Run command of show run on Switch0 and Switch1 and take screenshot of it.
- Verify SSH and do assign IP to another interface of Router. It should be done through laptop0. Take screen shot of it.



Task-03 Muhammad Nadeem Ghouri, a networking instructor at FAST University, assigned two hands-on tasks to his students to reinforce their skills in SSH and Telnet configurations on Cisco routers and switches. In the first task, students need to configure Telnet on a Cisco switch in Lab 08, where the university's networking team requires remote access for maintenance. They must enable VTY lines, set a login password, and verify Telnet access from a PC using command-line utilities. In the second task, students are required to configure SSH on a Cisco router in Lab 09 to ensure secure remote access for the university's IT administrators. They need to enable SSH version 2, generate RSA keys, create a local user account, and restrict remote access to SSH only. After completing the configuration, they must test the setup by connecting remotely from a PC using an SSH client.

Task-04 Muhammad Nadeem Ghouri, the network administrator of Tech Innovators Software House, is tasked with setting up remote access for the IT team to manage network devices securely. The software house has two departments: Development and IT Support, with a total of four PCs, one router, and one switch for internal connectivity. The IT Support team requires SSH access for secure remote management, while the Development team needs Telnet access for basic troubleshooting.

Network Setup:

1 Cisco Router (Router1)

1 Cisco Switch (Switch1)

4 PCs

PC1 & PC2 (Development Team) – Require Telnet access to Switch1

PC3 & PC4 (IT Support Team) – Require SSH access to Router1

Task Instructions:

Configure Telnet on Switch1

Assign a hostname: Switch1

Set an enable password: cisco123

Set up VTY lines (0-4) with a password for Telnet.

Allow only Telnet access on VTY lines.

Verify Telnet access from PC1 or PC2 using the telnet command.

Configure SSH on Router1

Assign a hostname: Router1

Set an enable secret: admin@123

Create a local user with privilege 15: Username: admin | Password: sshsecure

Generate RSA keys (1024-bit or higher) for SSH.

Enable SSH version 2 and restrict remote login to SSH only.

Verify SSH access from PC3 or PC4 using an SSH client (ssh admin@Router1).

Final Verification:

Ensure that PC1 & PC2 can connect to Switch1 using Telnet.

Ensure that PC3 & PC4 can securely connect to Router1 using SSH.

Use the show users and show ip ssh commands to verify active sessions.